Information Analysis and Visualisation

Tutorial Session 4

Data Processing Techniques

This laboratory session is focused on a range of data processing techniques. We will be using the Python libraries NumPy and pandas, which provides two key data structures, namely arrays and data frames. In pandas, the term Data Frames describes two-dimensional data structure organised and presented as a table. In many cases in practice, one of the first steps to perform when working with data is the so-called Data Cleaning, which usually includes removing or editing empty cells, unnecessary duplicates, incorrect formats and values.

Consider the following 28-row dataset, available on Moodle:

	longitude	latitude	housing_median_age	total_rooms	population	median_income	median_house_value	ocean_proximity
0	-122.23	37.88	41.0000	880	322	8.3252	452600	NEAR BAY
1	-122.23	37.88	41.0000	880	322	8.3252	452600	NEAR BAY
2	-122.22	37.86	21.0000	7099	2401	8.3014	358500	NEAR BAY
3	-122.25	37.84	52.0001	3104	1157	3.1200	241400	NEAR BAY
4	-122.26	37.85	52.0000	3503	1504	3.2705	241800	NEAR BAY
5	-121.65	39.32	40.0000	812	374	2.7891	73500	INLAND
6	-121.69	39.36	29.0000	2220	1170	2.3224	56200	INLAND
7	-121.70	39.37	32.0000	1852	911	1.7885	57000	INLAND
8	-121.70	39.36	46.0000	1210	523	1.9100	63900	INLAND
9	-121.70	39.36	37.0000	2330	1505	2.0474	116000	INLAND
10	-121.69	39.36	34.0000	842	635	1.8355	93000	INLAND
11	-121.74	39.38	27.0000	2596	1100	2.3243	85500	NaN
12	-121.80	39.33	30.0000	1019	501	2.5259	81300	INLAND
13	-120.46	38.15	16.0000	4221	1516	2.3816	81100	INLAND
14	-120.55	38.12	10.0000	1566	785	2.5000	81000	INLAND
15	-120.56	38.09	34.0000	2745	1150	2.3654	94900	INLAND
16	-124.23	41.75	11.0000	3159	1343	2.4805	73200	NEAR OCEAN
17	-124.21	41.77	17.0000	3461	1947	2.5795	68400	NEAR O
18	-124.19	41.78	15.0000	3140	1645	1.6654	74600	NEAR O
19	-124.16	41.74	15.0000	2715	1532	2.1829	69500	NEAR OCEAN
20	-124.14	41.95	21.0000	2696	1208	NaN	122400	NEAR OCEAN
21	-124.16	41.92	19.0000	1668	841	2.1336	75000	NEAR OCEAN
22	-118.32	33.35	27.0000	1675	744	2.1579	450000	ISLAND
23	-118.33	33.34	52.0000	2359	1100	2.8333	414700	ISLAND
24	-118.32	33.33	52.0000	2127	733	3.3906	300000	ISLAND
25	-118.32	33.34	52.0000	996	341	2.7361	450000	ISLAND
26	-118.48	33.43	29.0000	716	422	2.6042	287500	ISLAND
27	-118.48	33.43	29.0000	716	422	2.6042	287500	ISLAND

Actual dataset has approximately 20,000 rows and can be found at: https://www.kaggle.com/

Task 1

Download the Market.csv dataset from Moodle, load its content into pandas data frame and visualise the entire content of the data frame.

Task 2

Check the dataset for the following 'data cleaning' issues and resolve them:

- **Empty data**
- **Unnecessary duplicates**
- **Wrong formats**
- **Wrong values**

Task 3

The house prices are at the focus of this dataset. By using the new data frame, provide the following values which describe the column 'median_house_value':

- **Mean (average)**
- **Median**
- **Range, this is the difference between the maximum and the minimum values**

Task 4

Extract the column 'median_house_value' into a NumPy array. Visualise the array by using Line Chart, Box Plot Chart and Bar Chart. Determine which type of chart is the most suitable for visualising the data.